

Evaluation of the Use of the P-Model in Generating Synthetic Time Series of Solar and Tokamak Plasmas

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Abstract

The P-model, a stochastic model for multiplicative cascades, is widely used to study complex systems with intermittent fluctuations. We assess its use in generating synthetic time series of plasma dynamics from two reference datasets: (i) solar plasma fluctuations during a flare (AIA/SDO), and (ii) Tokamak emissivity simulations (SPARC→ITER, via NIMROD). Synthetic series were produced by tuning the asymmetry parameter p , and their statistical and spectral properties compared with those of the references. Lower p values better capture peaks in Tokamak data, while higher p values approximate solar plasma patterns. The model sometimes underestimates entropy and long-range correlations, revealing limitations. Overall, the P-model proves useful across plasma regimes and shows potential for generating synthetic datasets for plasma physics research.

1 Introduction

In various scenarios involving the training of artificial intelligence (AI) models to solve real-world problems, the availability of large volumes of data is essential. However, it is not always possible to obtain datasets large enough for such tasks, whether due to access restrictions, collection costs, or even a lack of representative data. This limitation can directly affect the quality of the trained model, as the scarcity of examples may compromise its generalization capacity [1, 9]. Therefore, it becomes necessary to seek complementary strategies to mitigate the effects of this scarcity.

One alternative is the generation of synthetic data capable of adequately representing the phenomena of interest. In plasma research, the availability of experimental data is often limited. In such cases, the use of mathematical models that simulate realistic data is essential to enable analyses and experiments. Another complementary approach, which has been gaining ground in recent years, is the use of generative artificial intelligence techniques, such as Generative Adversarial Networks (GANs), for producing synthetic time series [4].

Plasmas, in general, exhibit non-homogeneous characteristics due to the presence of instabilities, which can generate abrupt events. In the case of the Sun, such events include, for example, the emission of flares; in tokamaks, phenomena such as sudden discharges or plasma confinement loss may occur. Considering this, this article then presents an evaluation of the use of the P-model, originally applied to energy dissipation in turbulent systems, for generating synthetic time series of plasmas.

2 Methodology

2.1 P-Model

The P-model, proposed by Meneveau and Sreenivasan [8], simulates non-homogeneous cascades with multifractal fluctuations. It is based on the asymmetric redistribution of energy at each step of the cascade, producing time series with abrupt variations and intermittent behavior, also observed in plasmas.

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Multifractal in nature, it allows the simulation of endogenous or exogenous extreme events by adjusting the asymmetry parameter (p_1), with low computational cost, and theoretically corresponds to a generalization of a two-scale Cantor set. Deviations from the homogeneous cascade ($p_1 = p_2 = 0.5$) produce abrupt changes in frequency and magnitude, called extreme events (XE), classified as XE_{endo} or XE_{exo} by the SDGA formalism [10], typically occurring for $0.3 \leq p_1 < 0.5$ and $0.10 \leq p_1 < 0.3$, respectively (Fig. 1).

The asymmetry and intermittency properties of the P-model reflect collective and nonlinear processes in plasmas, making it useful for understanding spatial and temporal fluctuations. Techniques such as multifractal cascades and entropy measures [2, 3] appear promising for modeling this complexity, enabling the exploration of extreme event dynamics and the underlying multifractal structure in turbulent systems and plasmas.

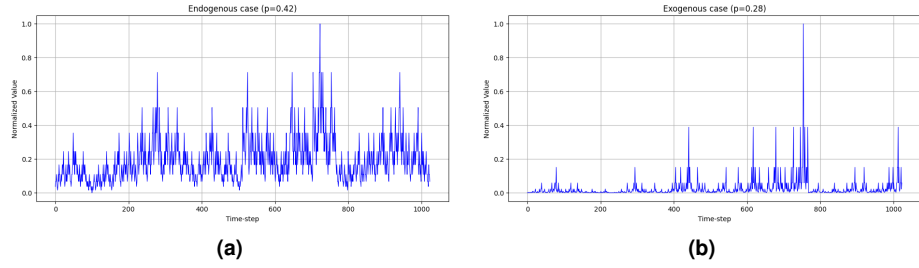


Figure 1. Examples of time series generated by the P-model: (a) endogenous extreme event (XE_{endo}) with $p = 0.42$ and (b) exogenous extreme event (XE_{exo}) with $p = 0.28$.

2.2 Real Data

2.2.1 Solar Dynamics Observatory (SDO) Data

To validate the P-model, a time series of a solar eruptive phenomenon observed by the Atmospheric Imaging Assembly (AIA) instrument on board the SDO satellite in the 304 Å band was used [5]. Although active region 12786 was relevant, the analysis focused on a rare and highly disruptive eruption at the solar limb, characterized by the abrupt emergence of a flux tube from a flux rope. The relevance of this event in the context of this work lies in its highly fluctuating temporal structure, with patterns similar to multiplicative cascades. In particular, we observed that the evolution of the maximum emission values in the region of interest exhibits intermittent and abrupt behaviors similar to synthetic XE_{exo} series simulated with the P-model. The time series was constructed from the average of the 10 highest intensity values in each of the 317 consecutive temporal windows of 10 seconds. The target region is highlighted in the first image of Figure 2 (yellow rectangle). The full video of the image sequence is available as supplementary material in the official GitHub repository [6].

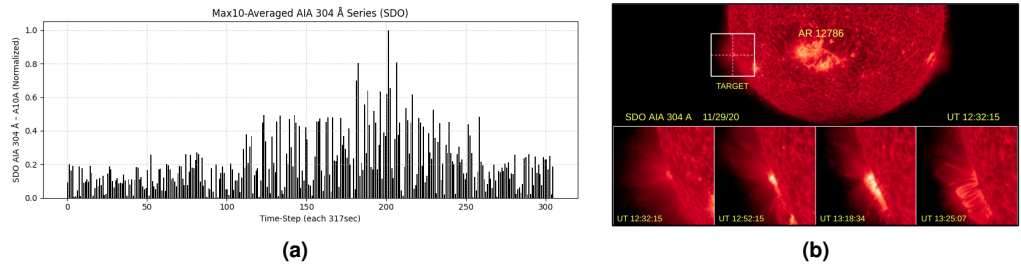


Figure 2. Selected SDO/AIA frames showing the solar eruption. (a) Time series of SDO/AIA 304 Å – A10A (normalized) from the yellow-marked region. (b) Snapshots of the corresponding solar region.

2.2.2 SPARC-ITER Tokamak Data

To generate plasma time series, the same methodology applied to the solar data was used for the SPARC-ITER Tokamak simulations. The data were obtained using the NIMROD code [7], adapted for ITER conditions and originally developed in the context of the SPARC project.

Figure 3 shows three frames from a high-resolution, non-resonant plasma emissivity simulation, tracking the evolution up to the peak radiated power. The third frame (at 3.05 ms) highlights the characteristic structure of a disruption, revealing the instability in the plasma. From this simulation, a representative time series of the heat flux (thermal emission) was extracted, effectively capturing the dynamic evolution of the system up to the disruptive event.

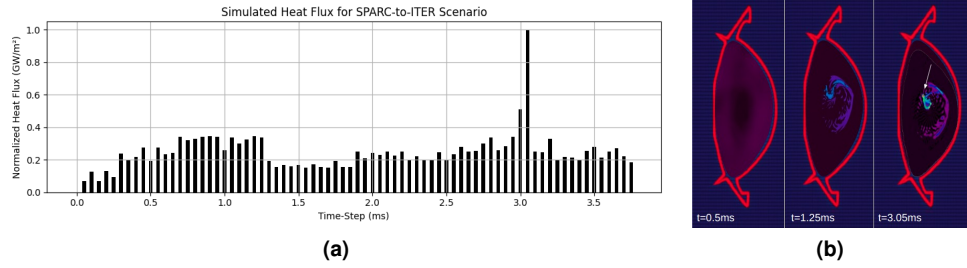


Figure 3. Snapshots from the SPARC-to-ITER simulation illustrating plasma behavior at 3.05 ms. (a) Time series of the heat flux extracted for analysis. (b) Plasma snapshots showing disruption and dynamics within the tokamak confinement.

2.3 Evaluation/Comparison Procedure

The ability of the P-model to reproduce statistical features of plasma time series, both in tokamaks and solar data, was evaluated through a comparative analysis between real and synthetic series generated for different values of the parameter p .

The comparison used metrics that capture higher-order statistical properties, spectral characteristics, and long-range correlations. The metrics employed were: skewness, kurtosis, entropy (a measure of series complexity), Kullback–Leibler (KL) divergence between the distributions of the real and simulated series, Detrended Fluctuation Analysis (DFA), and Power Spectral Density (PSD). For the skewness, kurtosis, entropy, DFA, and PSD metrics, the absolute difference between the values of the real and simulated series was calculated, while for the KL metric, the calculated value was used directly, interpreting lower values as indicative of a closer match between the distributions.

This methodology allows determining the value of p that best balances the evaluated metrics and analyzing the sensitivity of each metric with respect to variations in this parameter. Furthermore, it enabled discussion of the trade-offs between metrics, recognizing that different aspects of the time series may be better represented by different p values, although there are ranges in which the overall pattern is satisfactorily captured. The source code used during this study is available in the official project repository [6].

3 Results and Discussion

In the SDO data, the P-model exhibited p -dependent behavior. For $p < 0.3$, skewness and kurtosis were high, reflecting asymmetric fluctuations and heavy tails typical of exogenous events (XE_{exo}) (Figure 4). As p increases, these metrics approach real values, indicating that more endogenous regimes better reproduce solar plasma statistics. Entropy shows a similar trend—lower for $p < 0.3$, and closer to real data at higher p , though differences persist. KL divergence was lowest between $p = 0.25$ and 0.4 , suggesting better approximation of real distributions. DFA revealed an underestimation of long-range correlations, slightly improved for exogenous p ; similarly, the PSD showed smaller discrepancies for more exogenous regimes.

For the Tokamak, the P-model skewness best matched the real data for p between 0.2 and 0.3, while kurtosis showed higher similarity for p between 0.15 and 0.25, indicating a predominance of exogenous features (Figure 5). Entropy also approached real values for $p \sim 0.25$. KL divergence remained high but tended to decrease for exogenous p .

Concerning temporal dependencies, DFA indicated good reproduction of long-range correlations for p between 0.20 and 0.25, whereas PSD showed better agreement with real data for more endogenous p , highlighting a limitation of the model in this aspect.

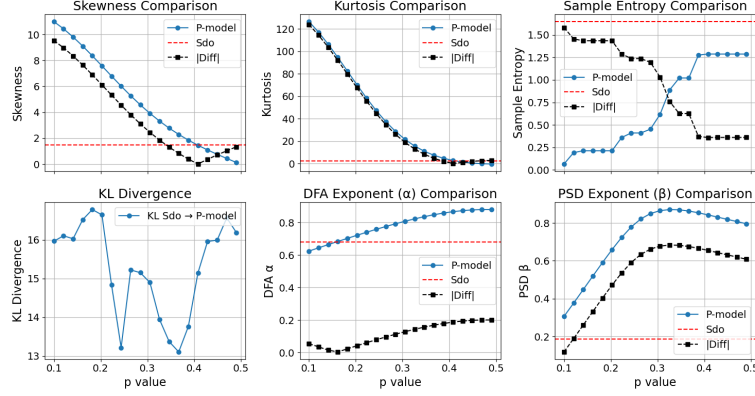


Figure 4. Comparison between SDO data (red line) and simulations (blue line) for different values of p . Metrics: (a) Skewness, (b) Kurtosis, (c) Entropy, (d) KL divergence, (e) DFA, and (f) PSD.

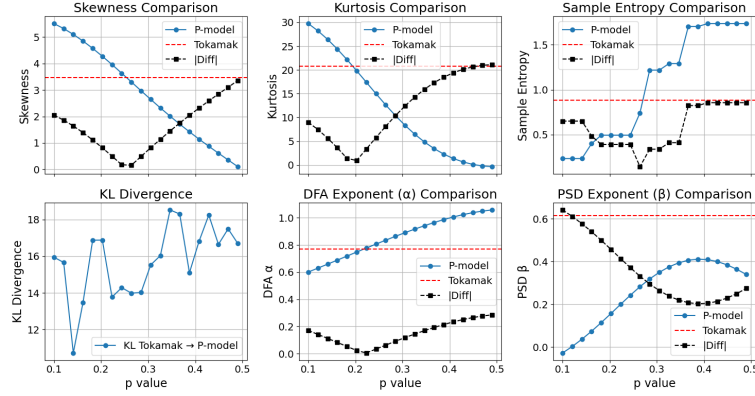


Figure 5. Comparison between real data (red line) and simulations (blue line) for different values of p . Metrics: (a) Skewness, (b) Kurtosis, (c) Entropy, (d) KL divergence, (e) DFA, and (f) PSD.

4 Conclusion

The results show that the P-model reproduces essential aspects of fluctuations in plasma time series, particularly extreme events and amplitude distributions. The best fit occurs for $p \approx 0.15!-!0.25$ in the Tokamak and $p \approx 0.25!-!0.4$ in the SDO data. However, the model has limitations, such as underestimation of entropy, partial reproduction of long-range correlations, and inconsistencies between metrics, making it difficult for a single value of p to simultaneously capture all statistical properties.

The use of the P-model is recommended as a fast generator of synthetic series for AI model training, combined with strategies such as p ensembles or hybridization with models that reproduce longer temporal dependencies. Future research should include optimization of p across multiple metrics, evaluation in different plasma regimes, and analysis of the impact of synthetic data on the performance of extreme event forecasting pipelines.

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